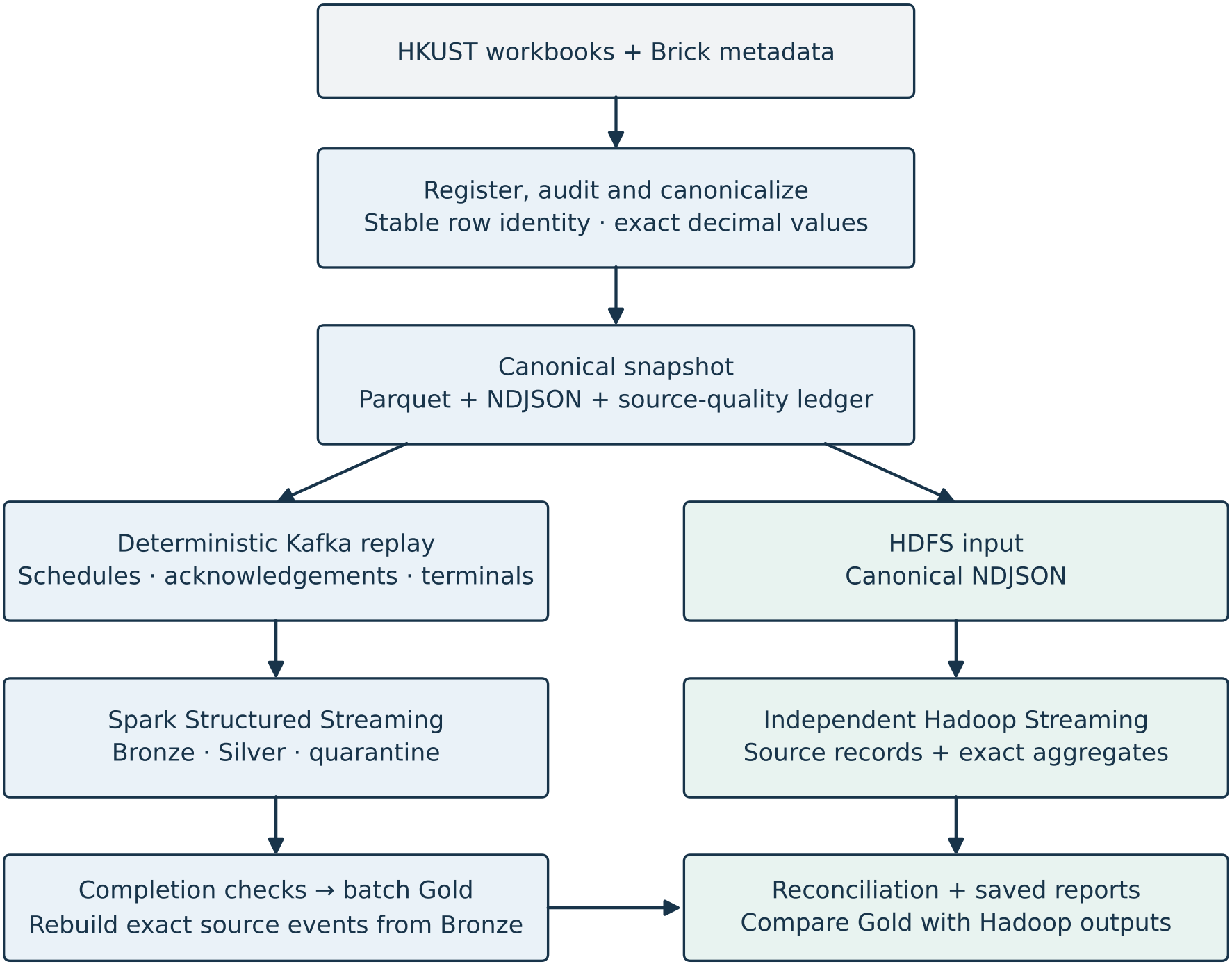
HKUST workbooks + Brick metadata
Register, audit and canonicalize
Stable row identity · exact decimal values
Canonical snapshot
Parquet + NDJSON + source-quality ledger
Deterministic Kafka replay
Schedules · acknowledgements · terminals
HDFS input
Canonical NDJSON
Spark Structured Streaming
Bronze · Silver · quarantine
Independent Hadoop Streaming
Source records + exact aggregates
Completion checks → batch Gold
Rebuild exact source events from Bronze
Reconciliation + saved reports
Compare Gold with Hadoop outputs
Read-only dashboard: delivery accounting, comparison coverage and recorded costs